

# EVERY $n$ POINTS IN THE PLANE CONTAIN $(m_1 + o(1))n$ POINTS WITH NO UNIT DISTANCE

SHISHENG LI

**ABSTRACT.** Let  $f(n)$  be the largest integer such that every set of  $n$  points in the plane contains  $f(n)$  points no two at distance 1 (Erdős Problem #1070). Let  $m_1$  be the supremum of the upper densities of measurable planar sets with no two points at distance 1. Larman and Rogers showed  $f(n) \geq m_1 n$ . We prove the matching asymptotic upper bound

$$f(n) = (m_1 + o(1))n,$$

which is the first half of Conjecture 1 of Dúcz and Varga. Combined with the bound  $m_1 \leq 0.247$  of Ambrus, Csiszárík, Matolcsi, Varga and Zsámboki, this gives  $f(n) \leq (0.247 + o(1))n$ . The proof transfers an arbitrary (non-measurable) independent set to a measurable one, using the Haar rigidity theorem for wild character laws from the OpenAI preprint *The Euclidean plane is not five-colorable* (2026); that theorem has a machine-checked Lean proof, and the whole argument is formalized in Lean on top of it.

## 1. INTRODUCTION

For a finite set  $P \subset \mathbb{R}^2$  let  $G_P$  be its unit-distance graph and  $\alpha(P)$  its independence number. Erdős asked to estimate

$$f(n) = \min_{P \subset \mathbb{R}^2, |P|=n} \alpha(P),$$

and in particular whether  $f(n) \geq n/4$ . The second question was answered negatively by Dúcz and Varga [3], who constructed a finite unit-distance graph with independence ratio below  $1/4$ . For the first, the forum discussion of the problem proposed  $f(n) = (m_1 + o(1))n$  as a resolution, and Dúcz and Varga state as their Conjecture 1 that  $f(n)/n = m_1(\mathbb{R}^2) + o(1)$  and that  $m_1(\mathbb{R}^2) = \delta_{\text{Croft}}$ . We prove the first half.

**Definition 1.1.** For measurable  $A \subset \mathbb{R}^2$  let

$$\bar{\delta}(A) = \limsup_{m \rightarrow \infty} \frac{|A \cap Q_m|}{|Q_m|},$$

where  $Q_m$  is either the ball  $B(0, m)$  or the square  $[-m, m]^2$ ; every argument below works for both. Let  $m_1$  be the supremum of  $\bar{\delta}(A)$  over measurable  $A$  containing no two points at distance 1, and

$$\alpha^* = \inf \left\{ \frac{\alpha(P)}{|P|} : P \subset \mathbb{R}^2 \text{ finite, nonempty} \right\}.$$

**Theorem 1.2.** (1)  $f(n) \geq \alpha^* n$  for every  $n$ , and  $f(n)/n \rightarrow \alpha^*$ .

(2)  $\alpha^* \geq m_1$  (Larman–Rogers [4]).

(3)  $m_1 \geq \alpha^*$ .

Hence  $m_1 n \leq f(n) \leq m_1 n + o(n)$ .

**Corollary 1.3.** By [1],  $m_1 \leq 0.247$ ; by Croft [2],  $m_1 \geq 0.22936$ . Hence  $0.22936 n \leq f(n) \leq (0.247 + o(1))n$ .

Parts (1) and (2) are elementary (Sections 2 and 3). Part (3) is the content of the paper (Section 4). Its only external input is Theorem 4.1 below, quoted from [6].

*Remark 1.4* (Status of the dependency). Theorem 4.1 is proved in the unreferenced preprint [6]. It is also proved in Lean 4 with Mathlib as the declaration `OAI.PlaneFiveColor.Spectral.wild_fourier` in [7], in exactly the Fourier form used here. We rebuilt that development (Lean v4.34.1, Mathlib d13f23b); `#print axioms` reports only `propext`, `Classical.choice`, `Quot.sound`. The whole argument of this paper is formalized in Lean on top of it (<https://github.com/daizisheng/erdos-1070>); the main theorems `Erdos1070.f_div_tendsto_m1` and `Erdos1070.m1_mul_le_f` also depend only on these three axioms. The Lean definition of upper density uses the squares  $[-m, m]^2$ .

## 2. CONVERGENCE OF $f(n)/n$

*Proof of Theorem 1.2(1).* If  $|P| = n$  then  $\alpha(P) \geq \alpha^*|P|$ , so  $f(n) \geq \alpha^*n$ .

Fix a finite  $P_0$  with  $|P_0| = s$ ,  $\alpha(P_0) = a$ . For  $N \geq s$  write  $N = ks + r$  with  $0 \leq r < s$ . Place  $k$  translates of  $P_0$  and  $r$  further single points so that distinct pieces are more than  $\text{diam}(P_0) + 2$  apart. The union  $P$  has  $N$  points and  $G_P$  is the disjoint union of the pieces' graphs, so

$$f(N) \leq \alpha(P) = ka + r \leq \frac{N}{s}a + s.$$

Thus  $\limsup_N f(N)/N \leq a/s$ ; taking the infimum over  $P_0$  gives  $\limsup f(N)/N \leq \alpha^*$ .  $\square$

## 3. LARMAN–ROGERS: $\alpha^* \geq m_1$

*Proof of Theorem 1.2(2).* Let  $A$  be measurable and unit-distance-free with  $\bar{\delta}(A) = \delta$ , and let  $P = \{p_1, \dots, p_n\}$ ,  $\rho = \max_j |p_j|$ . For every  $x$ ,  $(P + x) \cap A$  is independent in  $G_{P+x} \cong G_P$ . By Tonelli,

$$\int_{Q_m} |(P + x) \cap A| dx = \sum_j |A \cap (Q_m + p_j)| \geq \sum_j (|A \cap Q_m| - |Q_m \setminus (Q_m + p_j)|).$$

For balls and squares,  $|Q_m \setminus (Q_m + p)| \leq 2\pi m|p|$ , and  $|Q_m| \geq \pi m^2$ . Hence

$$\frac{1}{|Q_m|} \int_{Q_m} |(P + x) \cap A| dx \geq n \frac{|A \cap Q_m|}{|Q_m|} - \frac{2n\rho}{m}.$$

Along a subsequence on which the density ratio tends to  $\delta$ , the right side exceeds  $n\delta - \varepsilon$ , so some  $x$  has  $\alpha(P) \geq |(P + x) \cap A| > n\delta - \varepsilon$ . Hence  $\alpha(P) \geq m_1|P|$ .  $\square$

## 4. THE TRANSFER: $m_1 \geq \alpha^*$

**4.1. Setting and the cited theorem.** Identify  $\mathbb{R}^2$  with  $\mathbb{C}$ . Let  $F = \overline{\mathbb{Q}} \cap \mathbb{R}$ ,  $E = F(i)$  (the algebraic numbers) and  $K = \{u \in E : u\bar{u} = 1\}$ , both regarded as discrete groups. Let  $D = \widehat{E_{\text{disc}}}$  be the compact dual of  $E$ , with  $(ad)(z) = d(az)$  and  $\psi_z(d) = d(z)$  for  $a, z \in E$ . The continuous characters form the Borel subgroup  $C = j(\mathbb{R}^2)$ ,  $j(\xi)(z) = e^{i\xi \cdot z}$ . For  $x, y \in E$ ,  $|x - y| = 1$  iff  $x - y \in K$ .

**Theorem 4.1** ([6], Thm. 2.3; Lean: `wild_fourier`). *If  $\nu$  is a  $K$ -invariant probability measure on  $D$  with  $\nu(C) = 0$ , then  $\int_D \psi_z d\nu = 0$  for every  $z \in E \setminus \{0\}$ .*

We also use the following standard lemma, whose proof in [6, Lemma 4.1] we have checked line by line (spectral theorem for  $E_{\text{disc}}$ , Bochner's theorem on  $\mathbb{R}^2$ , Fourier uniqueness on  $D$ ).

**Lemma 4.2** ([6], Lemma 4.1). *Let a standard probability space carry a measure-preserving action of  $E$  with Koopman operators  $T_z$ . Then  $\mathcal{H}_c = \{v \in L^2 : \lim_{z \rightarrow 0, z \in E} \|T_z v - v\|_2 = 0\}$  is  $L^2$  of a factor  $\mathcal{F}_c$  and is the range of the spectral projection on  $C$ . On  $\mathcal{H}_c$  the  $T_z$  extend to a strongly continuous unitary representation  $T_x$ ,  $x \in \mathbb{R}^2$ , preserving the positive cone and the constant 1. If rotations  $V_u$  ( $u \in K$ ) satisfy  $V_u T_z V_u^{-1} = T_{uz}$ , then  $\mathcal{H}_c$  and its projection are rotation invariant.*

**4.2. An invariant random independent set with density at least  $\alpha^*$ .** Let  $\Omega \subset \{0, 1\}^E$  be the set of independent sets of the unit-distance graph on  $E$ . It is closed in the compact metrizable space  $\{0, 1\}^E$ . Translations  $s_z(\omega) = \omega - z$  and rotations  $\rho_u(\omega) = u\omega$  are homeomorphisms of  $\Omega$  with  $s_w \rho_u = \rho_u s_{u^{-1}w}$ .

**Lemma 4.3.** *There is an  $E \rtimes K$ -invariant Borel probability  $\mu$  on  $\Omega$  with  $\mu(0 \in \omega) \geq \alpha^*$ .*

*Proof.* One cannot simply average a single configuration over Følner sets of  $E \rtimes K$ : the visit counts are then non-uniform and one obtains a *weighted* independence ratio, which may be smaller than  $\alpha^*$ . We average in two stages.

*Translations.*  $E$  is countable abelian, so it has Følner sets  $F_N$ . Each  $F_N$  is a finite planar set; pick a maximum independent  $J_N \subset F_N$ , so  $|J_N| \geq \alpha^* |F_N|$ . Let  $\nu_N = |F_N|^{-1} \sum_{z \in F_N} \delta_{J_N - z}$ . Then  $\nu_N(0 \in \omega) = |J_N|/|F_N| \geq \alpha^*$  and  $\|(s_w)_* \nu_N - \nu_N\|_{TV} \leq |F_N \Delta (F_N + w)|/|F_N| \rightarrow 0$ . A weak-\* limit point  $\mu_0$  is  $E$ -invariant, and since  $\{0 \in \omega\}$  is clopen,  $\mu_0(0 \in \omega) \geq \alpha^*$ .

*Rotations.* With Følner sets  $G_M \subset K$ , let  $\mu_M = |G_M|^{-1} \sum_{u \in G_M} (\rho_u)_* \mu_0$ . Each  $(\rho_u)_* \mu_0$  is  $E$ -invariant because  $(s_w)_* (\rho_u)_* = (\rho_u)_* (s_{u^{-1}w})_*$ , and  $\mu_M(0 \in \omega) = \mu_0(0 \in \omega)$  because  $0 \in u\omega \iff 0 \in \omega$ . A weak-\* limit  $\mu$  is  $K$ - and  $E$ -invariant with  $\mu(0 \in \omega) \geq \alpha^*$ .  $\square$

**4.3. Projection to the continuous factor.** On  $L^2(\Omega, \mu)$  put  $T_z F(\omega) = F(\omega - z)$  and  $V_u F(\omega) = F(u^{-1}\omega)$ ; these are unitary,  $V_u T_z V_u^{-1} = T_{uz}$ , and  $f = \mathbf{1}[0 \in \omega]$  satisfies  $T_z f = \mathbf{1}[z \in \omega]$ ,  $V_u f = f$ . Since  $\omega$  is independent,  $f \cdot T_u f = 0$  for  $u \in K$ , so

$$(1) \quad \langle f, T_u f \rangle = 0 \quad (u \in K).$$

Let  $p = \mathbb{E}(f \mid \mathcal{F}_c)$  and  $b = f - p$ . Then  $0 \leq p \leq 1$ ,  $\mathbb{E}p = \mathbb{E}f \geq \alpha^*$ , and  $V_u p = p$ ,  $V_u b = b$  (Lemma 4.2).

The spectral measure  $\lambda_b$  of  $b$  is  $K$ -invariant:  $\langle b, T_{uz} b \rangle = \langle V_u^{-1} b, T_z V_u^{-1} b \rangle = \langle b, T_z b \rangle$ , and Fourier coefficients determine finite measures on  $D$ . Since  $b \perp \mathcal{H}_c$ ,  $\lambda_b(C) = 0$ . By Theorem 4.1 (applied to  $\lambda_b/\|b\|^2$  if  $b \neq 0$ ),  $\langle b, T_z b \rangle = 0$  for  $z \neq 0$ . As  $\mathcal{H}_c^\perp$  is  $T$ -invariant,  $\langle p, T_u b \rangle = \langle b, T_u p \rangle = 0$ . With (1),

$$\int p T_u p d\mu = 0 \quad (u \in K).$$

$K$  is dense in the unit circle (e.g.  $((1 - t^2) + 2ti)/(1 + t^2)$ ,  $t \in \mathbb{Q}$ ) and  $x \mapsto T_x p$  is continuous, so by unitarity

$$(2) \quad \int (T_x p)(T_{x+u} p) d\mu = 0 \quad (x \in \mathbb{R}^2, |u| = 1),$$

with  $0 \leq T_x p \leq 1$  and  $\mathbb{E}[T_x p] = \mathbb{E}p$ .

**4.4. A measurable field and one good sample.** Since  $x \mapsto T_x p$  is continuous into the separable space  $L^2(\mu)$ , there is a jointly measurable  $H : \mathbb{R}^2 \times \Omega \rightarrow [0, 1]$  with  $H(x, \cdot) = T_x p$  for  $x$  outside a null set. By Tonelli and (2), for every  $m$ ,

$$\int_{\Omega} \int_{B(0, m)} \int_{S^1} H(x, \omega) H(x + u, \omega) d\sigma(u) dx d\mu(\omega) = 0,$$

so a conull set  $\Omega_1$  of  $\omega$  has all inner integrals zero. Let  $A_m(\omega) = |Q_m|^{-1} \int_{Q_m} H(x, \omega) dx \in [0, 1]$ ; then  $\mathbb{E}A_m = \mathbb{E}p \geq \alpha^*$ , and by the reverse Fatou lemma  $X = \limsup_m A_m$  satisfies  $\mathbb{E}X \geq \alpha^*$ , hence  $\mu(X \geq \alpha^*) > 0$ . Fix  $\omega \in \Omega_1 \cap \{X \geq \alpha^*\}$  and  $h = H(\cdot, \omega)$ :

$$(3) \quad \int_{B(0,m)} \int_{S^1} h(x)h(x+u) d\sigma dx = 0 \quad \forall m, \quad \limsup_m \frac{1}{|Q_m|} \int_{Q_m} h \geq \alpha^*.$$

**4.5. Density points.** Let  $A_0 = \{h > 0\}$ . By (3),  $\int_{B(0,m)} \int_{S^1} \mathbf{1}_{A_0}(x) \mathbf{1}_{A_0}(x+u) d\sigma dx = 0$  for all  $m$ , and  $\bar{\delta}(A_0) \geq \alpha^*$  since  $h \leq \mathbf{1}_{A_0}$ . Let  $A$  be the set of Lebesgue density points of  $A_0$ ;  $A$  is Borel and  $|A \triangle A_0| = 0$ , so  $\bar{\delta}(A) \geq \alpha^*$ .

**Lemma 4.4.** *A contains no two points at distance 1.*

*Proof.* Suppose  $x, y = x + u \in A$  with  $|u| = 1$ . For small  $h_0 > 0$  let  $x'$  be uniform on  $B(x, h_0)$  and, independently,  $u'$  uniform on the arc  $\{e^{i\theta}u : |\theta| < h_0/2\}$ . Then  $\mathbb{P}(x' \notin A_0) \rightarrow 0$ , and, since conditionally on  $u'$  the point  $x' + u'$  is uniform on  $B(x + u', h_0) \subset B(y, 2h_0)$ ,  $\mathbb{P}(x' + u' \notin A_0) \leq 4|B(y, 2h_0) \setminus A_0|/|B(y, 2h_0)| \rightarrow 0$ . So  $\mathbb{P}(x' \in A_0, x' + u' \in A_0) > 0$ . The law of  $(x', u')$  has bounded density with respect to  $dx \otimes \sigma$  on  $B(0, m) \times S^1$  for  $m > |x| + 1$ , contradicting (3).  $\square$

Hence  $A$  is a measurable unit-distance-free set with  $\bar{\delta}(A) \geq \alpha^*$ , i.e.  $m_1 \geq \alpha^*$ . This proves Theorem 1.2(3) and the theorem.

#### ACKNOWLEDGEMENTS

The negative answer to the second question of #1070 is due to Dúcz and Varga [3]. The rigidity theorem and the transfer template are from [6]; the present argument is a single-label version of its Proposition 4.2.

#### REFERENCES

- [1] G. Ambrus, A. Csiszárík, M. Matolcsi, D. Varga, P. Zsámboki, *The density of planar sets avoiding unit distances*, Math. Program. (2023); arXiv:2207.14179.
- [2] H. T. Croft, *Incidence incidents*, Eureka **30** (1967), 22–26.
- [3] Á. Dúcz, D. Varga, *A unit-distance graph in the plane with independence ratio below 1/4*, arXiv:2606.28157 (2026).
- [4] D. G. Larman, C. A. Rogers, *The realization of distances within sets in Euclidean space*, Mathematika **19** (1972), 1–24.
- [5] M. Matolcsi, I. Z. Ruzsa, D. Varga, P. Zsámboki, *The fractional chromatic number of the plane is at least 4*, arXiv:2311.10069.
- [6] OpenAI, *The Euclidean plane is not five-colorable*, preprint, 23 September 2026, <https://github.com/openai/math>.
- [7] OpenAI, `openai/math` Lean development, commit `adc7f12`, files `OAI/Analysis/PlaneSpectrum/*`.